



Short- and Medium-term Prognosis in Patients Hospitalized for COPD Exacerbation

The CODEX Index

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Background: No valid tools exist for evaluating the prognosis in the short and medium term after hospital discharge of patients with COPD. Our hypothesis was that a new index based on the CODEX (comorbidity, obstruction, dyspnea, and previous severe exacerbations) index can accurately predict mortality, hospital readmission, and their combination for the period from 3 months to 1 year after discharge in patients hospitalized for COPD.

Methods: A multicenter study of patients hospitalized for COPD exacerbations was used to develop the CODEX index, and a different patient cohort was used for validation. Comorbidity was measured using the age-adjusted Charlson index, whereas dyspnea, obstruction, and severe exacerbations were calculated according to BODEX (BMI, airflow obstruction, dyspnea, and previous severe exacerbations) thresholds. Information about mortality and readmissions for COPD or other causes was collected at 3 and 12 months after hospital discharge.

Results: Two sets of 606 and 377 patients were included in the development and validation cohorts, respectively. The CODEX index was associated with mortality at 3 months ($P < .0001$; hazard ratio [HR], 1.5; 95% CI, 1.2-1.8) and 1 year ($P < .0001$; HR, 1.3; 95% CI, 1.2-1.5), hospital readmissions in the same periods, and their combination (all $P < .0001$). All CODEX C statistics were superior to those of the BODEX, DOSE (dyspnea, airflow obstruction, smoking status, and exacerbation frequency), and updated ADO (age, dyspnea, and airflow obstruction) indexes.

Conclusions: The CODEX index was a useful predictor of survival and readmission at both 3 months and 1 year after hospital discharge for a COPD exacerbation, with a prognostic capacity superior to other previously published indexes.

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Abbreviations: ADO = age, dyspnea, and airflow obstruction; AECOPD = acute exacerbation of COPD; AUC = area under the curve; BODE = BMI, airflow obstruction, dyspnea, and exercise capacity; BODEX = BMI, airflow obstruction, dyspnea, and previous severe exacerbations; CODEX = comorbidity, obstruction, dyspnea, and previous severe exacerbations; DOSE = dyspnea, airflow obstruction, smoking status, and exacerbation frequency; ESMI = EPOC en los Servicios de Medicina Interna; HR = hazard ratio; mMRC = modified Medical Research Council; ROC = receiver operating characteristic

COPD is one of the most prevalent diseases in adults and is associated with high morbidity and mortality worldwide.¹ Although many variables are associated with increased mortality in patients with COPD, the most frequently recognized are postbronchodilator FEV₁, dyspnea, and exacerbations.^{2,3} Of interest, all these variables have been incorporated into the newly released GOLD (Global Initiative for Chronic Obstructive Lung Disease) recommendations for grading COPD

severity.⁴ The use of multidimensional indexes, such as the BODE (BMI, airflow obstruction, dyspnea, and exercise capacity), BODEX (BMI, airflow obstruction,

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dyspnea, and previous severe exacerbations), ADO (age, dyspnea, and airflow obstruction), DOSE (dyspnea, airflow obstruction, smoking status, and exacerbation

frequency), and other indexes, have improved the prognostic capacity beyond individual variables.⁵⁻⁹ However, to our knowledge, none has demonstrated prognostic value in the short term after a severe acute exacerbation of COPD (AECOPD), such as after a hospital discharge, which suggests that their usefulness may vary depending on the length of the study observation and the severity of disease.¹⁰

Although studies suggest that posthospital mortality may have declined, it continues to remain too high.¹¹ It is well known that hospitalizations due to COPD are still associated with poorer survival in the months following, and this is at least in part because many patients are older, with advanced respiratory disease and frequent comorbidities. These three characteristics are all well-recognized indicators of mortality in overall COPD populations.^{12,13}

Interestingly, comorbidities measured with the Charlson index are related to global but not to respiratory mortality in the original BODE index cohort. However, the original BODE index cohort was recruited from among ambulatory patients without significant comorbidities.⁵ A more recent study performed by the same group concluded that selected comorbidities are associated with higher mortality from COPD- and non-COPD-related causes, although, again, in this work, comorbidity was not related to survival in the short or medium term.¹⁴ Conversely, two other studies performed in patients hospitalized for AECOPD related comorbidity to mortality and hospital readmission at 90 days after discharge.^{15,16} It is precisely in this group of patients, with higher mortality in the short-term follow-up, that prognosis reliability is most needed, although regrettably none of the current indexes has proved to be useful.

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The main objective of our study was to develop and validate a new index to predict short-term mortality and readmissions in patients hospitalized for a severe acute COPD exacerbation. For this purpose, we selected five variables with previously recognized usefulness in the prognostic evaluation of these patients (comorbidity, age, obstruction, dyspnea, and exacerbations), using previously validated cutoff points whenever possible.

MATERIALS AND METHODS

Data from the EPOC en los Servicios de Medicina Interna (ESMI) study were used for the development of the CODEX (comorbidity, obstruction, dyspnea, and previous severe exacerbations) index. The ESMI study is a longitudinal, observational, multicenter study conducted in 70 EDs and internal medicine services in Spain. Investigators included the first 10 consecutive patients seen for a severe COPD exacerbation during a 1-year period (October 2009–October 2010). Full methodology is available elsewhere and is summarized in e-Appendix 1.^{16,17} During admission, all patients were evaluated using a standardized questionnaire. Comorbidity was documented using the previously validated Charlson index, a standard scale with 15 chronic diseases graded for severity of disease, in which one point is added to the total score for each decade of life over the age of 50 years.¹⁸ Dyspnea was measured using the modified Medical Research Council (mMRC) scale. The vital status and number of hospital readmissions for COPD or other causes at 3 months and 1 year after hospital discharge were collected at outpatient visits and, when this was not possible, by reviewing medical records or through telephone calls to the patient or family. Respiratory and other causes of death were recorded by the responsible investigator. Dyspnea, obstruction measured with postbronchodilator FEV₁, expressed as a percentage, and the number of hospitalizations for AECOPDs during the previous year were stratified according to previously published BODE and BODEX thresholds.^{5,6} Finally, the original Charlson index was stratified in tertiles (0–4 = 0; 5–7 = 1; ≥ 8 = 2). As a result, possible scores on the newly proposed CODEX index range from 0 to 10 points (Table 1).

The validation cohort analysis was performed by pooling data from three previously published cohorts of patients hospitalized at University Hospital Mutua de Terrassa, all with the same inclusion and exclusion criteria as in the ESMI study.^{11,16,19} Additionally, we evaluated the usefulness of the CODEX index in predicting the combined end point of mortality or readmission for any cause, both at 3 and 12 months, and compared its usefulness with the BODEX, DOSE, and updated ADO indexes to predict mortality, readmissions, and their combination in both time periods.

The ESMI study and the studies used for the validation cohort were approved by the Clinical Investigation Ethics Committee of the Hospital Mútua de Terrassa (EO/0922_0709). All patients participated voluntarily and signed written informed consent forms.

Statistical Analysis

Qualitative variables were expressed as absolute frequencies and percentages, and quantitative variables were summarized as mean and SD in the case of normally distributed, or as median otherwise. Comparisons among means were made using the Student *t* test or Mann-Whitney *U* test for continuous variables and the χ^2 or Fisher exact test for categorical variables. Time-dependent variables from hospital discharge were analyzed with Cox logistic regression and Kaplan-Meier statistics. To assess the ability of the

Table 1—Variables and Thresholds to Estimate the CODEX Index

CODEX	Domain	Variables	Scoring			
			0	1	2	3
C	Comorbidity	Charlson index ^a	0-4	5-7	≥ 8	...
O	Obstruction	FEV ₁ %	≥ 65	50-64	36-49	≤ 35
D	Dyspnea	mMRC scale	0-1	2	3	4
EX	Exacerbation	Exacerbation ^b	0	1-2	≥ 3	...

CODEX = comorbidity, obstruction, dyspnea, and previous severe exacerbations; mMRC = modified Medical Research Council.

^aCharlson index: one point is added to the total score for each decade of life over the age of 50 y.

^bSevere exacerbations of COPD during the previous year (hospitalization or ED consultations).

CODEX index to predict all-cause mortality or readmissions, and to make comparisons with other mortality indexes, we plotted receiver operating characteristic (ROC) curves, and their areas under the curve (AUCs) were compared with C statistics. Comparative significance was assessed with the DeLong method.²⁰ In all comparisons, statistical significance was identified for *P* values < .05.

RESULTS

Development Cohort

Overall, 679 patients were screened, and 606 were included in the initial ESMI study. Fifteen patients were excluded for having an incomplete minimum dataset, and in another 58 cases, spirometry was unable to be performed or did not fulfill the spirometric criteria for COPD (Fig 1). No differences existed between included and excluded patients with respect to age, sex, smoking history, or dyspnea measured by the mMRC scale (data not shown). Our study included patients hospitalized for exacerbations of COPD, regardless of whether they required mechanical ventilation, either invasive or not.

The mean age of the ESMI study population was 72.6 years (SD, 9.9 years), and 89.8% were men. Sev-

enty patients (11.5%) were discharged from the ED without requiring conventional hospitalization. Patients discharged directly from the ED were younger, with a higher FEV₁ and fewer comorbidities (all *P* < .01). For this reason, we initially performed a subanalysis that showed no differences between the CODEX index and the mortality rate at 3 months, after excluding patients discharged from the ED. The rest of the analysis was conducted with the whole ESMI cohort.

Validation Cohort

The validation cohort consisted of 377 patients hospitalized in the University Hospital Mútua de Terrassa during three different time periods (1997-1998, 1999-2000, and 2003-2004). The average age of these patients was 71.9 years (SD, 9.7 years), with a predominance of men (94%). The main characteristics of the ESMI and validation cohorts are presented in Table 2.

The mean value for the CODEX index in hospitalized patients in the ESMI study was 4.9 (2.4), decreasing slightly to 4.7 (2.4) when patients discharged from the ED were incorporated (*P* = not significant), with a median of 5 points both in the ESMI study and in the validation cohort. Figure 2 shows the frequency point distribution of the CODEX index in the ESMI and validation cohorts (*P* = not significant).

Mortality

Vital status at 3 months was collected for 557 case patients (91.9%) in the ESMI study, 27 of whom died

Table 2—Demographic and Clinical Characteristics of the ESMI and Validation Cohorts

Variables	ESMI (n = 606)	Validation Cohort (n = 377)	<i>P</i> Value
Age, y	72.6 (9.9)	71.9 (9.4)	ns
Male sex, %	546 (90.1)	353 (94.0)	.01
BMI	27.5 (5.1)	26.9 (5.0)	ns
Length of stay, d	8.2 (7.7)	11.3 (7.6)	< .0001
Smoking history, No. (%)	572 (94.4)	350 (93.6)	.03
Packs/y	55.6 (27.9)	59.6 (35.6)	ns
Dyspnea mMRC ^a	2.3 (1.1)	2.5 (1.2)	ns
COPD hospitalizations ^b	1.6 (1.6)	1.3 (1.7)	.01
PCO ₂ at admission, mm Hg	47.5 (13.3)	43.1 (12.6)	< .001
PO ₂ , mm Hg	60.5 (16.6)	62.4 (18.5)	ns
Charlson index	3 (2)	2.1 (1.2)	< .001
Katz index	5.3 (1.2)	5.5 (1.1)	< .001
FVC % ^c	68.9 (19)	64.7 (20.7)	< .001
FEV ₁ % ^c	45.4 (15.6)	41.2 (14.8)	< .001
FEV ₁ /FVC	52 (12.7)	47.3 (10.8)	< .001
Cor pulmonale, No. (%)	143 (23.6)	126 (33.7)	< .05

Data are presented as mean (SD) unless indicated otherwise. ESMI = EPOC en los Servicios de Medicina Interna; ns = not significant. See Table 1 legend for expansion of other abbreviation.

^aDyspnea measured with mMRC scale.

^bNo. hospitalizations for COPD during previous year.

^cPostbronchodilator test.

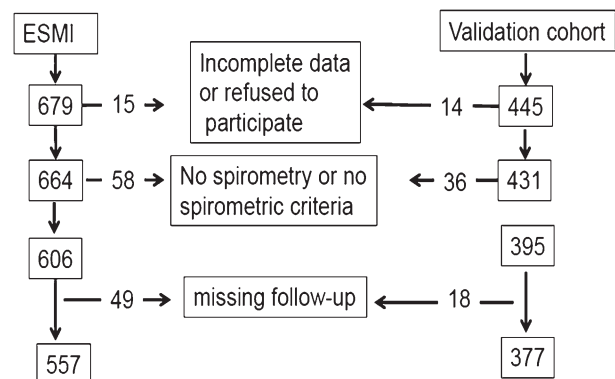


FIGURE 1. Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) flowchart of participants and causes of exclusion. ESMI = EPOC en los Servicios de Medicina Interna.

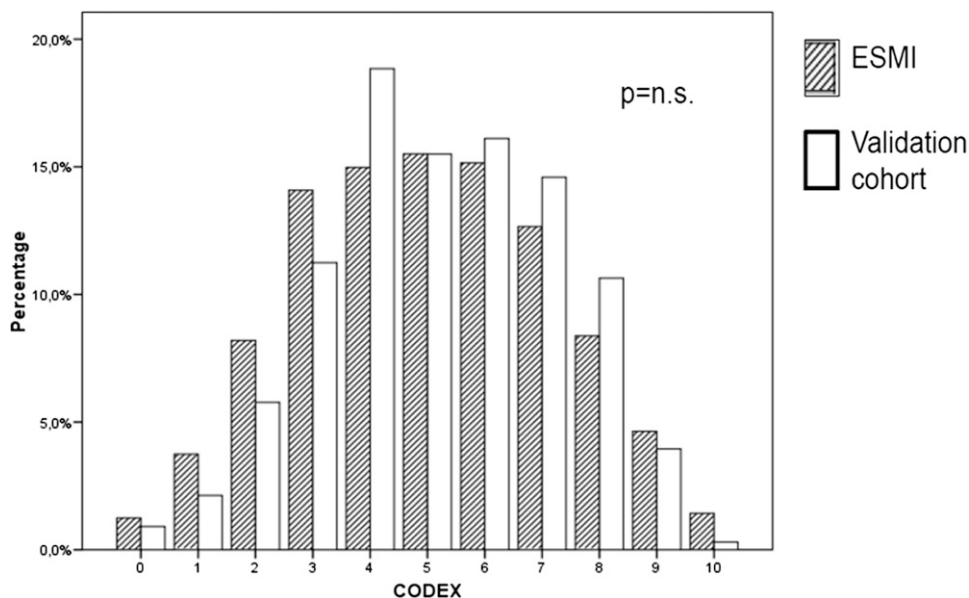


FIGURE 2. Distribution of the CODEX index in the ESMI and validation cohorts. CODEX = comorbidity, obstruction, dyspnea, and previous severe exacerbations; n.s. = not significant. See Figure 1 legend for expansion of other abbreviations.

within 3 months after discharge (4.8%). Mortality at 3 months was associated with age, number of hospital admissions for COPD or other causes in the previous year, greater dyspnea (mMRC score), chronic home oxygen therapy use, more functional dependence (Katz index), and Charlson index both with and without age stratification, but was not associated with BMI ($P = .2$) (e-Table 1). The CODEX index was statistically associated with mortality ($P < .0001$; hazard ratio [HR] 1.5; 95% CI, 1.2-1.8). Mortality in the validation cohort was 7%, and the CODEX index showed a similar association ($P = .006$; HR, 1.4; 95% CI, 1.1-1.7). The AUC was 0.73 for the ESMI cohort, 0.71 for the validation cohort, and 0.72 for the combination of the two.

Mortality at 30 days was 2.1% in the ESMI study and 1.6% in the validation cohort. Vital status was assessed at 1 year in 932 patients of the ESMI study and validation cohort (95.1%), and of these, 126 (13.5%) had died. No differences were found between patients with and without complete follow-up in terms of age, sex, length of hospitalization, dyspnea, FEV₁, or score on the CODEX index (data not shown). Mortality causes were respiratory (58%), cardiovascular (16%), and neoplasm (5%); other or unknown causes accounted for 21%. The CODEX index was significantly related to mortality at 1 year in both the ESMI study ($P < .0001$; HR, 1.3; 95% CI, 1.2-1.5) and the validation cohort ($P < .0001$; HR, 1.4; 95% CI, 1.2-1.5). The CODEX index was also a significant predictor of mortality for respiratory causes at 1 year after discharge ($P < .0001$; HR, 1.28; 95% CI, 1.16-1.4).

The AUC at 1 year was 0.67 for the ESMI study and 0.68 for the validation cohort. Kaplan-Meier curves

for survival at 3 months and 1 year for the global population are represented graphically in Figure 3.

Combined Variable

The CODEX index was significantly related to hospital readmission at both 3 months (5.7 [2.2] vs 3.5 [3.3], $P < .0001$) and 1 year (5.5 [2] vs 4.7 [2.2], $P < .0001$). In assessing the combined variable (death or readmission), 34.2% of patients had died or had been readmitted for COPD at 3 months, increasing to 53.8% at 1 year. In both periods, the CODEX index was significantly related to the presence of the composite variable: 3 months (6.1 [2.2] vs 4.6 [2.2]) and 1 year (5.6 [2.1] vs 4.6 [2.1]) ($P < .0001$ for both). Figure 4 shows the relationship between the different outcome variables and the scores obtained on the CODEX index.

Comparison With Other Indexes

The updated ADO, DOSE, and BODEX indexes were significantly related to the probability of death at 3 months and at 1 year after hospital discharge (Table 3). However, comparison of the ROC curves (C statistics) up to 3 months after discharge showed a better predictive capacity for the CODEX index compared with the BODEX ($P < .005$), DOSE ($P < .01$), and updated ADO ($P < .05$) indexes. This difference remained significant at 1 year for the BODEX ($P < .02$) and DOSE indexes ($P < .02$), but not for the updated ADO index (Fig 5).

The possibility of readmission at 3 months reached statistical significance for the BODEX (4.8 [2.5]

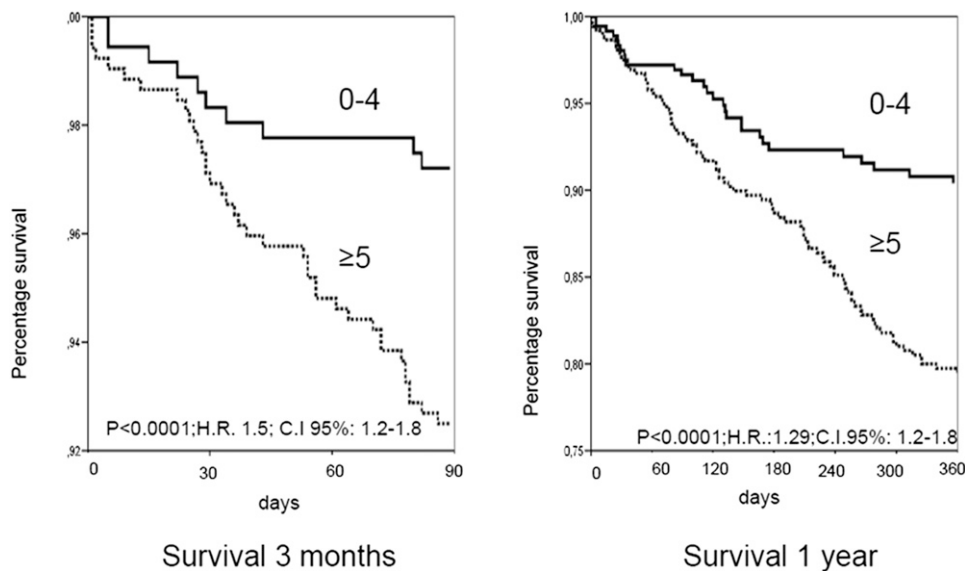


FIGURE 3. Kaplan-Meier curves for mortality. H.R. = hazard ratio.

vs 4.2 [2.1], $P = .008$) and DOSE (4.3 [1.7] vs 3.6 [1.8], $P > .01$) indexes, but not for the updated ADO index. These results are similar to those observed during the year after discharge: BODEX index ($P = .002$), DOSE index ($P < .001$), and updated ADO index ($P = .1$).

With respect to the combined variable (death and/or rehospitalization), all indexes reached statistical significance at 3 months and at 1 year. New ROC curves showed a better predictive capacity at 90 days from hospital discharge for the CODEX index compared with the BODEX ($P < .005$) and ADO ($P < .0001$) indexes, although no statistical difference was observed with the DOSE index. The 1-year analysis showed similar results for the comparison of the AUC in the ROC curves between the CODEX index and the ADO index ($P < .0001$), the BODEX index ($P < .001$), and the DOSE index ($P < .05$). Full comparison analysis

among the several indexes and outcomes is detailed in e-Tables 2 to 5.

DISCUSSION

Our study, performed in patients hospitalized for AECOPD, has three main findings. First, we demonstrated that a simple index can be useful in predicting survival in both the short and medium term after hospital discharge in these patients. Second, we reported the usefulness of this CODEX index in evaluating the risk of readmission, as well as the composite end point (readmission and/or mortality). Third, we had the opportunity to compare the prognostic validity with other previously validated indexes in the general population with COPD, and to compare their efficacy with the new index in patients with more severe disease, such as those requiring hospital admission for AECOPD. To our knowledge, no previous multicomponent, simple index has been shown to be useful in predicting mortality in a global population of patients admitted for AECOPD in periods shorter than 1 year after hospital discharge.¹⁰

Few studies have examined the short-term prognosis of patients hospitalized for COPD, and these were performed in selected populations, such as patients admitted to ICUs or those with hypercapnia at admission. In addition, the proposed prognostic indexes are excessively complex and are not applicable to routine clinical practice.²¹⁻²³ More recently, a new index was developed for predicting in-hospital mortality in patients with AECOPD, and it uses a different approach from that of our study. Of note, in this study, patients with comorbidities who were expected to have survival limited to < 12 months were excluded.²⁴

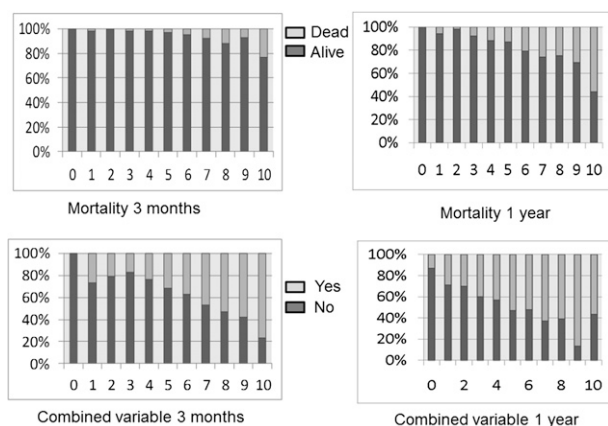


FIGURE 4. Scores on the CODEX index and mortality or combined variable. See Figure 2 legend for expansion of abbreviation.

Table 3—Prognostic Validity of Mortality (HR and AUC) at 3 Mo and 1 Y After Hospital Discharge of CODEX Index and All Other COPD Multivariate Indexes (ADO, BODEX, BODE, and DOSE) in the ESMI Study

Index	P Value	HR	95% CI	AUC	95% CI AUC
3 mo					
CODEX	.001	1.5	1.2-1.8	0.72	0.69-0.75
ADO	.001	1.3	1.1-1.6	0.65	0.58-0.65
BODEX	.006	1.1	1.04-1.3	0.62	0.58-0.65
DOSE	.005	1.3	1.1-1.5	0.6	0.57-0.63
1 y					
CODEX	.001	1.3	1.1-1.5	0.68	0.65-0.71
ADO	.001	1.2	1.1-1.4	0.64	0.61-0.67
BODEX	.001	1.3	1.1-1.4	0.51	0.48-0.55
DOSE	.001	1.3	1.1-1.5	0.59	0.56-0.63

ADO = age, dyspnea, and airflow obstruction; AUC = area under the curve; BODE = BMI, airflow obstruction, dyspnea, and exercise capacity; BODEX = BMI, airflow obstruction, dyspnea, and previous severe exacerbations; DOSE = dyspnea, airflow obstruction, smoking status, and exacerbation frequency; HR = hazard ratio. See Table 1 and 2 legends for expansion of other abbreviations.

The most widely recognized variables associated with decreased survival in COPD populations are lung function, measured with FEV₁ after bronchodilator test; dyspnea; and exacerbations. The first two are the

basis of multicomponent indexes. It is well established that FEV₁ is a good predictor of mortality in the general population, and historically it has been considered the most important variable for evaluating severity in

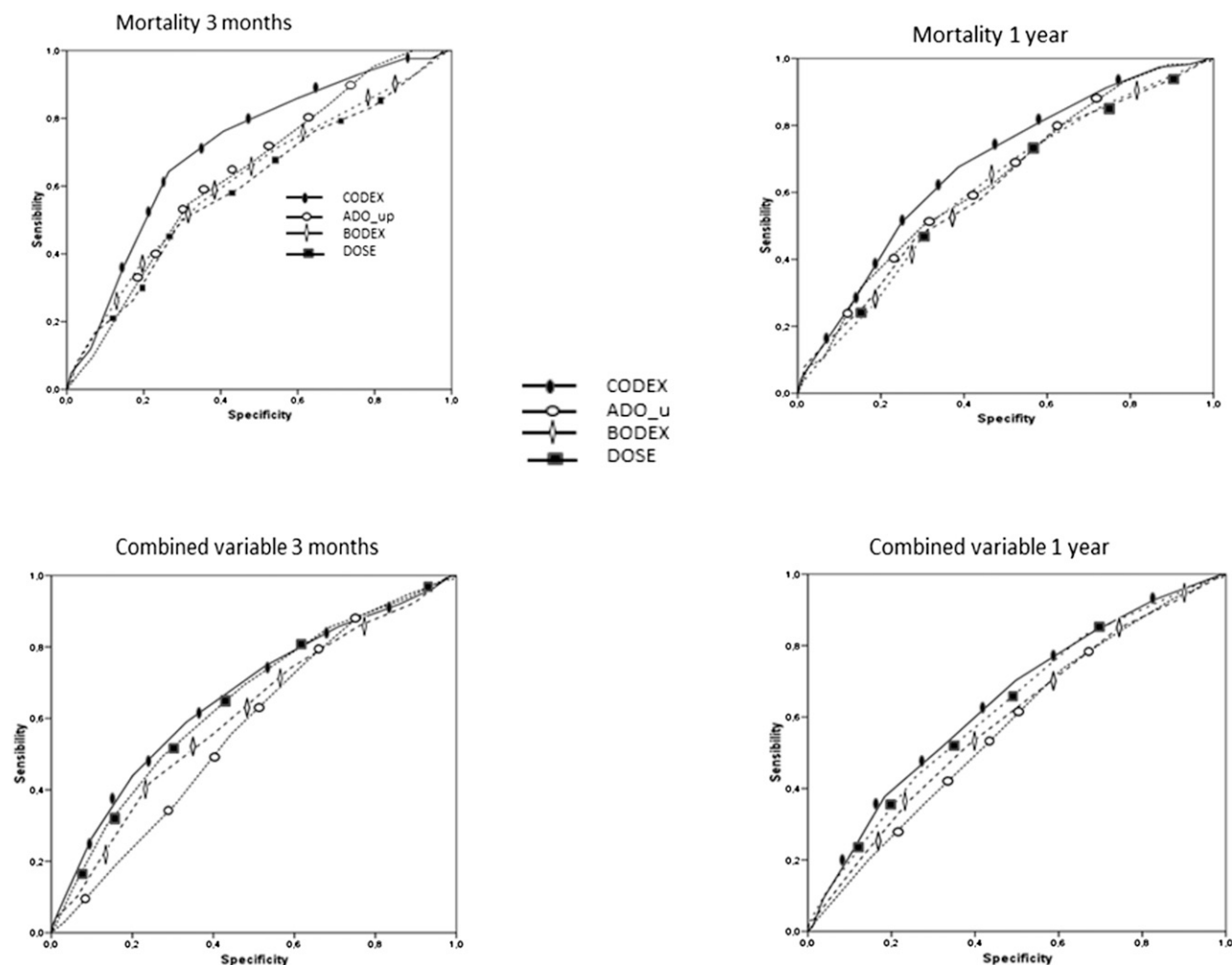


FIGURE 5. ROC curves for the CODEX, BODEX, DOSE, and updated ADO_u indexes. ROC = receiver operating characteristic. ADO_u = age, dyspnea, and airflow obstruction; BODEX = BMI, airflow obstruction, dyspnea, and previous severe exacerbations; DOSE = dyspnea, airflow obstruction, smoking status, and exacerbation frequency. See Figure 2 legend for expansion of other abbreviations.

COPD. Furthermore, the decline of FEV₁ over time has been used to indicate progression of the disease, although considerable individual variation exists.²⁵ The second variable, dyspnea, is the most frequent and limiting symptom in COPD, and its relationship with mortality is even more important than that observed for pulmonary function.² In contrast, the role of severe exacerbations in the natural history of COPD, especially those requiring hospitalization or an ED visit, has been recognized relatively recently.^{3,13} An increased frequency of exacerbations indicates clinical activity of the disease and is associated with a deterioration of comorbidities and worsening health status. The number of these severe exacerbations increases with the progression of the disease, especially in the advanced stages of COPD, and they are associated with a marked increase in mortality in the weeks following the exacerbations.²⁶

Multicomponent scales have been developed to improve prognosis prediction in COPD, and they have proved to be better predictors of survival than any isolated variable. The most recognized and widely used is the BODE index.⁵ This index has shown not only prognostic reliability in several studies, but also sensitivity to clinical changes such as the impact of exacerbations and improvement with rehabilitation.²⁷ However, the BODE index was developed and validated in outpatients without significant comorbidities,⁵ and, to our knowledge, only one study performed in a limited number of patients has proven its usefulness as a predictor of long-term mortality in patients hospitalized for COPD exacerbations.¹⁹ Alternatively, the BODEX index replaced the 6-min walk test component for exacerbation frequency a year ago and is more easily performed in this population, with a preserved power to predict mortality risk.⁶ Another recent multicomponent scale, the DOSE index, is associated with an increased risk of exacerbation in the following year, although it has been shown to be related to mortality at 5 years.^{9,28}

Finally, the ADO index, composed of age, FEV₁, and dyspnea stratified into different thresholds, has demonstrated its usefulness as a mortality predictor and has been updated to a larger cohort.^{7,8} However, the influence of age on the updated ADO index may appear clinically excessive (eg, being between 60 and 69 years of age has the same prognostic implication as presenting an FEV₁ < 35% and more than having the maximal degree of dyspnea in the mMRC scale). Obviously, age is a predictor of mortality in our study as well (e-Table 1). However, our intention was to build on already validated cutoffs (FEV₁, dyspnea, and exacerbations from the BODE/BODEX) and not to seek new cut points based exclusively on the criterion of statistical power. In this sense, to measure comorbidity, we used the most widely accepted comorbidity

index (Charlson index) in its original version (ie, with correction for age, adding one point for each decade from age 40 years). The Charlson index has shown its usefulness in survival prediction and health-care resource use in COPD and other chronic diseases.^{29,30} Of note, as reflected in e-Table 1, the prognostic significance of the Charlson index without age correction is very similar ($P < .0001$; HR, 1.35; 95% CI, 1.18-1.57) to the age-stratified Charlson index that was used in the CODEX index ($P < .0001$; HR, 1.26; 95% CI, 1.15-1.39). Additionally, the CODEX index is also related to respiratory mortality.

This newly proposed CODEX index is essentially an evolution of the BODE and BODEX indexes, retaining their cutoffs for dyspnea, obstruction, and previous exacerbations, but replacing BMI with comorbidity measured using the original Charlson index modified by age. Although it is well known that a low BMI is associated with a poor survival even in patients hospitalized for AECOPD,³¹ in our study, the Charlson index showed a better prognostic capacity. In addition, the prognostic reliability of the CODEX index does not improve by adding the BMI or functional dependence measured with the Katz index (data not shown).

The importance of comorbidities in the prognosis of patients with COPD is well established.⁴ Even in studies in which patients with severe comorbidities were excluded by design, a significant percentage of patients died of cardiovascular disease or cancer during follow-up.^{5,32,33} Studies performed in hospitalized patients with COPD have shown a strong independent association with comorbidity and medium- to long-term mortality after discharge.^{15,16,34} However, to date, none of the COPD indexes has included comorbidity as a prognostic variable. Divo et al¹⁴ reported the usefulness of a comorbidity scale in COPD in improving the prognostic capacity of the BODE index. Our results confirm that compared with other previous COPD multicomponent indexes, the CODEX index is the most useful in predicting survival, hospital readmissions, and a combination of the two in the short and medium term in patients hospitalized for AECOPD. Only the updated ADO index in the assessment of mortality at 1 year and the DOSE index in the combined variable at 3 months show a similar relationship between sensitivity and specificity.

Our study has several limitations. First, comparison with the BODE index was not possible because the 6-min walk test was not included in the ESMI study protocol, because performing the 6-min walk test can be difficult in this population. Second, our study had a clear predominance of men. This finding is similar to all studies of COPD performed in our area and is probably related to the fact that fewer Spanish women than men smoked in the 1940s and 1950s.³⁵ Finally,

the validation cohort comes from the same institution in different time periods, but with the same inclusion and exclusion criteria of the ESMI study.

CONCLUSIONS

In conclusion, the CODEX index is a simple, valid tool for predicting survival and risk of COPD readmission or their combination in patients hospitalized for AECOPD during the year after hospital discharge, and it can be used to assess the severity of COPD in this population. Determining its usefulness in other populations requires further study.

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Author contributions: Dr Almagro had full access to all the data in the study and takes responsibility for its integrity and the accuracy of the data analysis.

Dr Almagro: contributed to the study design, analysis and interpretation of the data, and writing of the manuscript.

Dr Soriano: contributed to the study design, analysis and interpretation of the data, and writing of the manuscript.

Dr Cabrera: contributed to the data collection and review of the manuscript.

Dr Boixeda: contributed to the data collection and review of the manuscript.

Dr Alonso-Ortiz: contributed to the data collection and review of the manuscript.

Dr Barreiro: contributed to the data collection and review of the manuscript.

Dr Díez-Manglano: contributed to the data collection and review of the manuscript.

Dr Murio: contributed to the study design and writing of the manuscript.

Dr Heredia: contributed to the data collection and review of the manuscript.

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Additional information: The e-Appendix and e-Tables can be found in the "Supplemental Materials" area of the online article.

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